

Prathamesh Bhavsar

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Summary

AI/ML Engineer with 4 years of experience designing, implementing, and deploying scalable AI/ML and GenAI solutions across enterprise environments. Strong experience in end-to-end ML project delivery, including business requirement gathering, data preparation, model development, deployment, monitoring, and stakeholder communication. Skilled in building production-grade AI systems, MLOps pipelines, cloud-based architectures, and customer-focused solutions using Python, SQL, AWS, LangChain, and modern ML frameworks.

Skills

- **Programming & Databases** - Python, SQL, Spark, Snowflake, MongoDB, NumPy, Pandas
- **AI/ML & GenAI** - TensorFlow, PyTorch, LangChain, GPT-4, BioBERT, RAG, Vector Search, Prompt Engineering
- **Cloud & MLOps** - AWS, SageMaker, Bedrock, EC2, S3, Step Functions, MLflow, Airflow, Docker, CI/CD
- **Solution Delivery** - Cloud Architecture, ML Pipelines, Model Deployment, Monitoring, Stakeholder Communication

Experience

Data and AI Co-Op, Tenet Healthcare - Dallas, TX Aug 2025 – May 2026

- Designed and delivered a production-grade **GenAI medical coding solution** using **LangChain and GPT-4**, translating stakeholder requirements into scalable AI workflows; improved coding accuracy from **78% to 89%** and reduced manual effort from **100 hours to 45 hours**.
- Built an end-to-end **clinical trial matching pipeline** using **BioBERT, graph-based retrieval, and ML workflows**, improving screening efficiency and increasing enrollment from **100 to 142 patients**.
- Operationalized a real-time **AI decision support system** with deployment, monitoring, and stakeholder-facing insights across **8 ICUs**, reducing mortality from **100 cases to 88 cases**.
- Collaborated with healthcare, data, and engineering teams to align **AI/ML solutions** with business objectives, compliance-sensitive workflows, and production delivery requirements.

Software Development Engineer (Machine Learning), EnelX - Mumbai, India Jul 2021 – Aug 2024

- Developed and deployed scalable **machine learning services** for predictive maintenance, supporting high-throughput data processing and reducing outages from **100 incidents to 60 incidents**.
- Designed an **AI-driven energy analytics platform** for business stakeholders, transforming enterprise data into actionable insights and reducing average energy costs from **\$100K to \$76K** across **300+ facilities**.
- Built distributed **data and ML pipelines** integrating streaming workflows and knowledge graphs, improving reliability and reducing root cause identification time from **2 hours to 35 minutes**.
- Created an optimization engine for **EV fleet operations**, operationalizing predictive models to improve cost efficiency and increase uptime from **90% to 97%** across **800+ vehicles**.
- Led cross-functional data analysis across **30+ suppliers**, improving data quality, reducing duplicate overhead from **12% to 5%**, and saving effort from **100 hours to 50 hours monthly**.
- Designed reusable backend services and **MLOps-ready data pipelines** processing millions of records daily, improving system efficiency from **70% to 90%**.

Education

University of Texas at Dallas, MS in Computer Engineering Aug 2024 – May 2026

University of Mumbai, Bachelor of Engineering in Information Technology Jul 2017 – Jun 2021

Projects

Doclytics (RAG, FAISS, Azure OpenAI, Streamlit) GitHub Link

- Built a **RAG-based GenAI application** using FAISS, Azure OpenAI, and Streamlit to extract structured insights from enterprise documents and support faster decision-making.

AgentToyota — Smart Vehicle Financing AI Agent (FastAPI, React, LangChain) GitHub Link

- Developed a full-stack **AI agent workflow** using FastAPI, React, and LangChain, enabling scalable decision automation and reducing decision time by **45%**.

StreamFraud: Real-Time Credit Card Fraud Detection (Kafka, Docker, Isolation Forest) GitHub Link

- Built a containerized **real-time ML pipeline** using Kafka and Docker for scalable anomaly detection, achieving **99.79% accuracy**.